🛡️ Fraud Detection Suite: Real-Time Risk & Intelligence

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Target Stakeholders: Fraud Operations, Risk Management, & Chief Risk Officers (CRO)

📊 Executive Summary

The **Fraud Detection Suite** is an executive-grade decisioning platform designed to mitigate financial leakage through real-time predictive scoring and geospatial transaction intelligence. By transforming raw transaction logs into actionable risk signals, the suite enables institutions to move from reactive mitigation to proactive prevention, directly supporting **NACHA 2026 Fraud Monitoring** requirements.

Strategic Business Value:

- **Operational Resilience:** Automates initial triage for high-velocity environments, targeting a **sub-100ms alert generation** benchmark to stop fraud at the point of sale.
- **Precision-Based Triage:** Leverages a Random Forest Classifier to reduce "over-blocking," aiming for a **false positive rate below 15%** to protect legitimate customer revenue.
- **Regulatory Readiness:** Built to meet Phase 1 (March 2026) and Phase 2 (June 2026) Nacha rule changes requiring risk-based processes to identify atypical transaction patterns.

🚀 Strategic Modules & Intelligence

The portal features a "Human-in-the-Loop" architecture designed for high-fidelity investigation and rapid case resolution.

🧠 Predictive Risk Engine

- **Goal:** Score every transaction for potential fraud using behavioral and financial telemetry.
- **Logic:** Utilizes a **Random Forest Classifier** trained on volume, merchant category, and "distance from home" outliers (calculated via the Haversine formula).
- **Versioned Health:** Tracks model drift over time to ensure detection baseline remains resilient against evolving fraud typologies.

📍 Geospatial Intelligence (GIS)

- **Goal:** Visualize the "where and when" of fraud to identify regional "hot zones."
- **Visualizations:** Integrates **PyDeck** with `ArcLayer` to map the physical displacement between a user's home base and the transaction origin.
- **Hot Zone Mapping:** Uses `HeatmapLayers` to identify geographic clusters of fraud, enabling regional-level policy adjustments.

⚖️ Case Management & Merchant Intel

- **Goal:** Facilitate deep-dive audits of suspicious accounts and merchants.
- **Analytics:** Benchmarks individual transactions against historical merchant averages and monitors "spending velocity" to differentiate genuine surges from fraudulent spikes.

🛠️ Technical Implementation

This project demonstrates proficiency in integrating complex machine learning pipelines with high-fidelity, interactive frontends.

Layer	Technology Stack
Frontend	Streamlit (Session state management & multi-tab navigation)
Intelligence	`Scikit-Learn` (Random Forest), `Pandas`, `NumPy`
Visualization	`PyDeck` (Mapbox integration), `Plotly Express` (Risk distribution)

📊 Strategic KPI Glossary

Metrics calculated within the suite are aligned with **2026 Executive Fraud Reporting** standards:

- **Fraud-to-Sales Ratio:** High-level financial exposure; benchmarked against an industry standard of **less than 1%**.
- **Precision Rate:** The ratio of fraudulent transactions to total declines, ensuring minimal customer friction.
- **Mean Time to Detect (MTTD):** Measures the agility of the system in identifying and flagging new fraud trends as they emerge.

💻 Installation & Access

To launch the Fraud Detection Portal locally:

1. **Clone the repository:** `git clone https://github.com/your-username/fraud-detection-suite.git`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Launch the application:** `streamlit run app.py`

Admin Gateway Credentials:

- **Username:** `admin`
- **Password:** `Risk2026`

📁 Repository Strategy

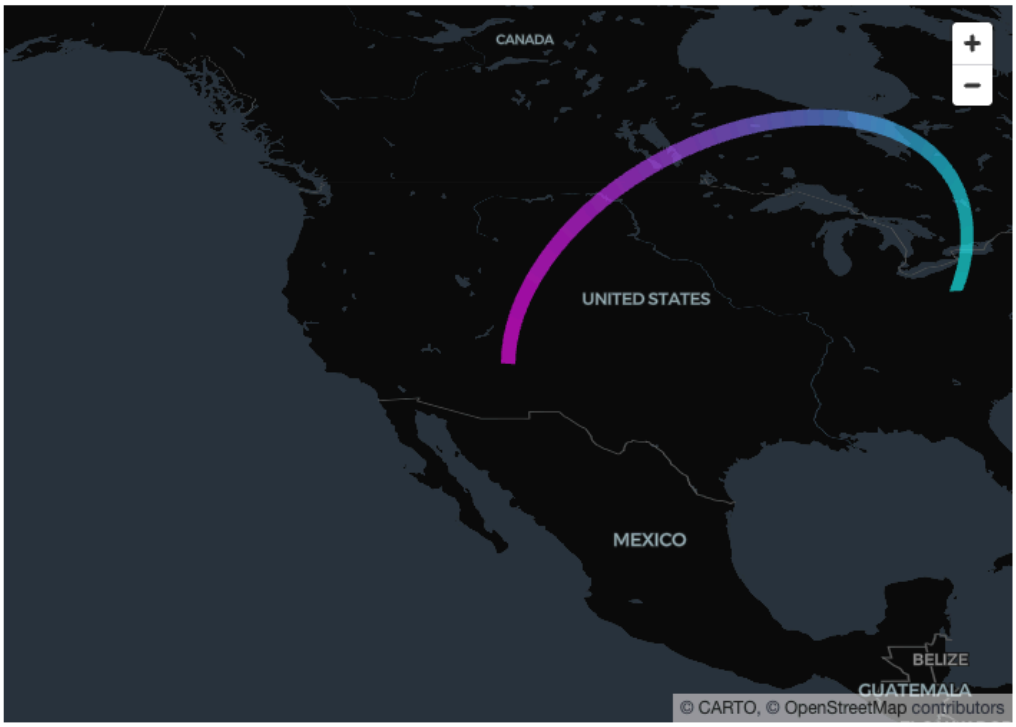
This repository serves as a **Narrative of Authority**, shifting the focus from simple monitoring to institutional resilience.

Operational Focus (Old)	Strategic Focus (New)
Lists "Geospatial Map" as a feature.	Frames it as "Geospatial Intelligence" for Day-0 detection.

Operational Focus (Old)	Strategic Focus (New)
Describes ML as "Random Forest."	Frames it as "Precision-Based Triage" to protect ROTCE.

Screen Shots

Case ID: 0 | Los Angeles, CA



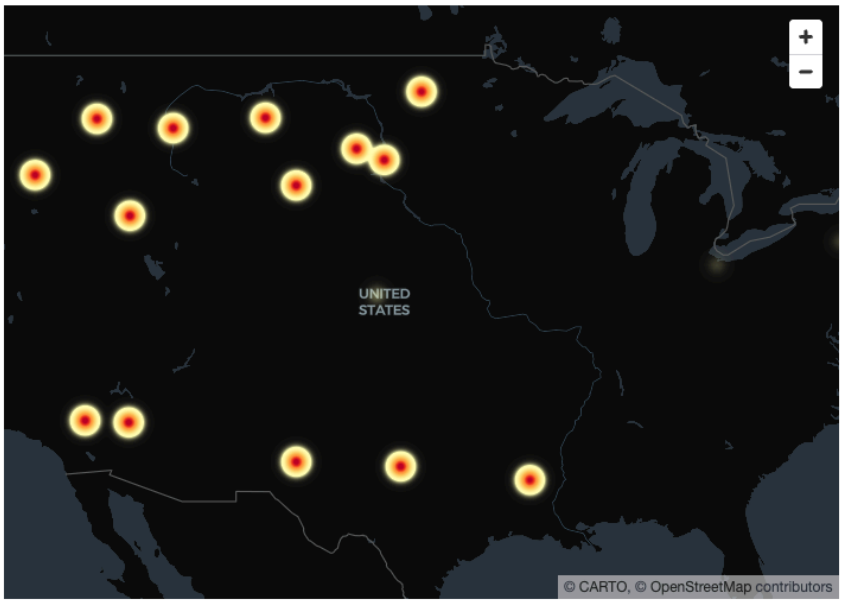
id	timestamp	merchant	Location	amount	risk_score	priority
165	2026-04-08 17:28:07.581062	Retail	Chicago, IL	265.84	0	Low
388	2026-04-08 15:14:07.581908	Gas	Miami, FL	658.37	0	Low
699	2026-04-08 13:58:07.583065	Grocery	Seattle, WA	1119.88	100	High
644	2026-04-08 13:35:07.582865	Retail	Seattle, WA	1078.71	100	High
906	2026-04-08 13:05:07.583838	Grocery	Miami, FL	956.02	0	Low
440	2026-04-08 10:42:07.582099	Grocery	Los Angeles, CA	889.18	100	High

Geographic Risk Density

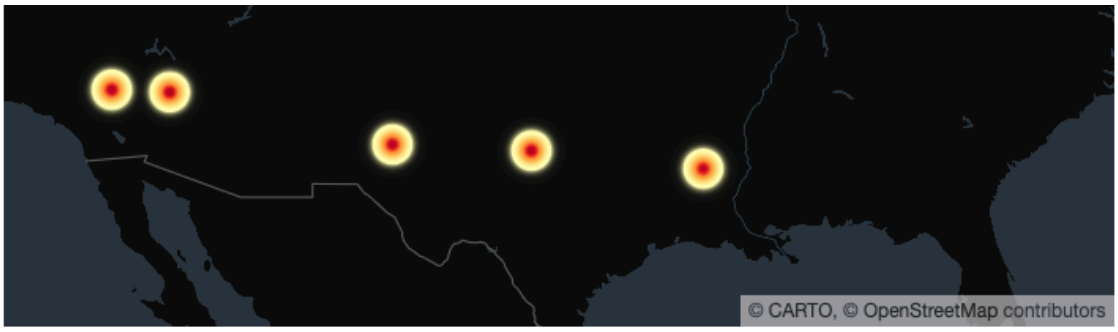
How to read this map: Heat clusters represent cumulative risk. Bright red areas indicate high-frequency fraud attempts occurring within the selected hour.

Filter by Hour (24h Format):

12



id	merchant	Location	amount	risk_score	priority
0	Retail	Los Angeles, CA	882.21	100	High



id	merchant	Location	amount	risk_score	priority
0	Retail	Los Angeles, CA	882.21	100	High
1	Retail	Miami, FL	769.13	98	High
2	Grocery	Seattle, WA	397.56	3	Low
3	Digital	Miami, FL	290.58	0	Low
4	Grocery	Miami, FL	949.72	100	High
5	Gas	Miami, FL	575.34	0	Low
6	Grocery	Seattle, WA	516.32	1	Low
7	Retail	Los Angeles, CA	538.50	0	Low
8	Digital	Miami, FL	531.94	0	Low
9	Grocery	Seattle, WA	478.40	0	Low

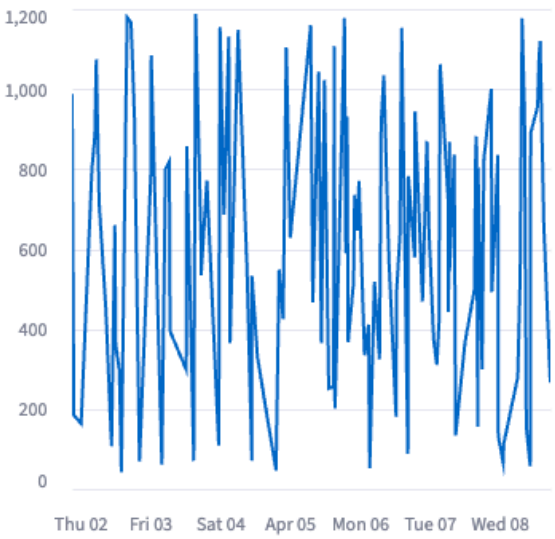
Fraud Detection Suite v1

Case Analysis | Global Heatmap | Merchant Intel | Model Health

Merchant Intelligence & Behavior Analysis

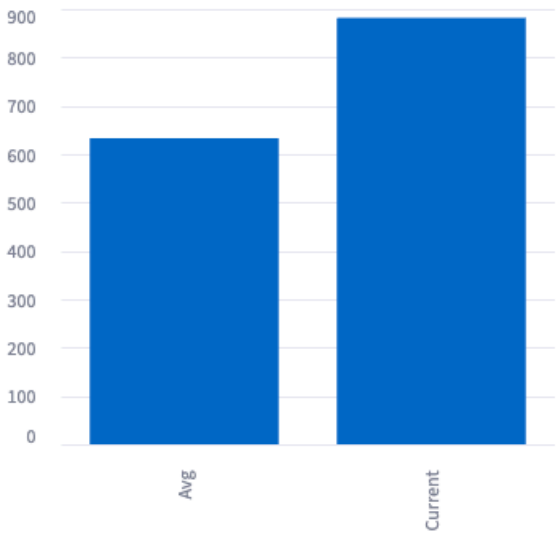
This tab provides a deep dive into individual user spending patterns and category-wide risk trends.

User 1005 Spending Velocity



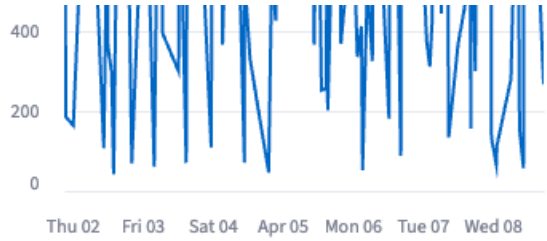
Velocity Analysis: Sudden vertical spikes signal outlier behaviors deviating from historical norms.

Merchant Context: Retail



Peer Benchmarking: Compares current amount against historical merchant average.

Rolling Merchant Risk Velocity (24h)



Velocity Analysis: Sudden vertical spikes signal outlier behaviors deviating from historical norms.

Peer Benchmarking: Compares current amount against historical merchant average.

🚀 Rolling Merchant Risk Velocity (24h)

📍 Filter Velocity by Geography:

Northeast

Merchant	Velocity	Current Avg Risk
Gas	0.26	0.6
Digital	-0.06	0.2
Grocery	-0.1	0
Retail	-0.16	0

💡 **Note:** Trends for **Northeast** region. Velocity > 5.0 indicates emerging risk clusters.

Fraud Detection Suite v1

Case Analysis | Global Heatmap | Merchant Intel | Model Health

System Performance & Auditor Oversight

Current Baseline Risk

41.31%

Model Version

v1

↑ Active

Retrain Model & Refresh Data

Version Comparison History

Version	Baseline Risk	Timestamp

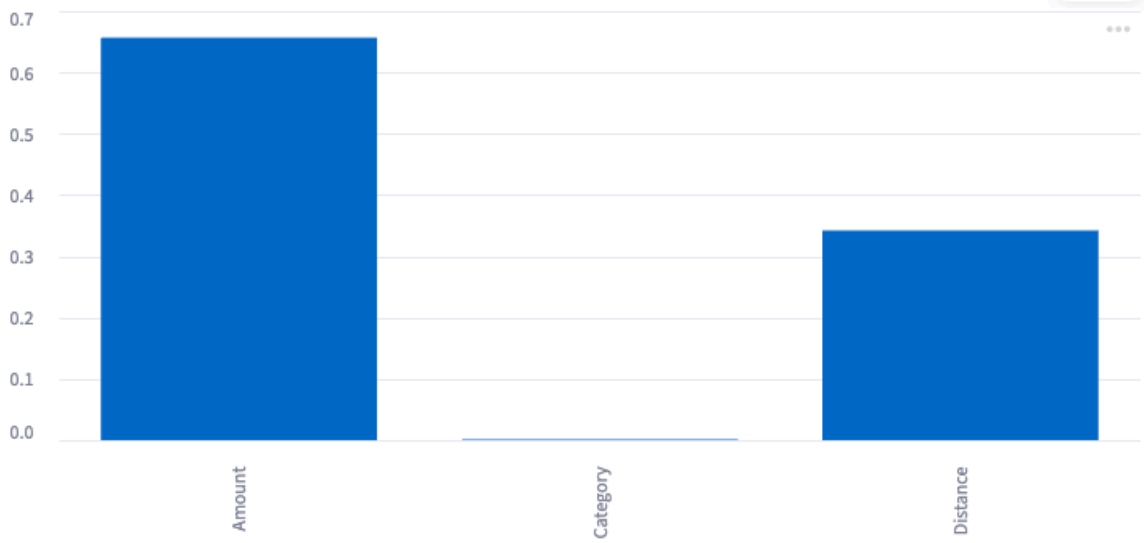
41.31%

v1

↑ Active

🔄 Retrain Model & Refresh Data

📜 Version Comparison History

Version	Baseline Risk	Timestamp
1	41.31%	17:50:07

Feature Importance